Simali Bouchra

FUTURE COMPUTER SCIENCE & NETWORKS ENGINEER — AI & DATA SPECIALIZATION

CONTACT INFORMATION

+212 674437322

simalibouchra20@gmail.com

Casablanca - Morocco

Simali Bouchra

SOFT SKILLS

- Teamwork
- Adaptability
- Problem-Solving
- Time Management & Organization
- Continuous Learning

TECHNICAL SKILLS

Programming:

- Python, Java, C, C++, C#, SQL, Bash

AI & Machine Learning:

- Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, TF-IDF, NLP

Data Engineering:

- ETL, Dimensional Modeling, SSIS/SSRS, Hadoop

Databases:

- MySQL, MongoDB, Oracle 19c (PL/SQL), SQL Server (T-SQL)

Development & Tools:

- Git, GitHub, Docker, Node.js, React, React Native

Web Development:

- HTML, CSS, JavaScript, Bootstrap

Modeling & Methodologies:

- UML, MERISE 2, Agile/Scrum

PROFESSIONAL EXPERIENCE

Predictive Maintenance system— JESA **01/07 -31/08 (2026)**

- Introduced a data fusion strategy to contextualize data across digital maintenance systems.
- Developed a System of Systems approach to unify interconnected maintenance systems.
- Built a predictive maintenance pipeline for failure prediction, root-cause diagnosis, and RUL estimation.
- Integrated Neo4j, MQTT, and Node-RED for contextualization and real-time monitoring.

LANGUAGES

- Anglais (Courant)
- Français (Courant)
- Arabe (Maternelle)

Interests

- Martial Arts
- Reading
- Gaming

PROFIL

AI & Data Engineering student pursuing an Engineering degree in Computer Science and Networks, with a specialization in AI & Data. Expected to graduate in 2027 and seeking a 6-month final-year project internship (PFE) starting in February 2027.

EDUCATION

Engineering Degree in Computer Science and Networks — AI/Data Option **2024- 2027**

Moroccan School of Engineering Sciences (EMSI), Casablanca, Morocco **Casablanca**

Preparatory Years — Engineering Studies **2022 - 2024**

Moroccan School of Engineering Sciences (EMSI), Casablanca, Morocco **Casablanca**

Baccalaureate in Experimental Sciences — Physics & Chemistry **2021 - 2022**

Groupe Scolaire Ambassadeur, Casablanca, Morocco **Casablanca**

Mention Bien

ACADEMIC PROJECTS (OTHERS AVAILABLE ON GITHUB)

Biblio Archivist : 28/10/2025- 21/12/2025

- Designed and developed a mobile application dedicated to the management and reading of digital books.
- Optimized user experience through smooth navigation, an interactive library, and detailed book browsing.
- Developed an enhanced reading module with visual customization and user profile integration for reading tracking and progress monitoring.
- Implemented advanced features including dynamic bookmarks, favorites management, and quick access to saved reading positions through the user profile.

Technologies utilisé : React Native, MongoDB, HTML, CSS, Bootstrap, JavaScript

Ad_On: Classified Ads Website 01/12/2024 - 01/02/2025

- Full-stack web development: real-time publishing, browsing, and management of classified listings.
- User profiles and integrated messaging: listing management and direct communication between buyers and sellers.
- Advanced search and filtering by category, keywords, price, location, and product type.
- Administration and security: listing moderation, user management, and protection of sensitive data.

Technologies utilisé : Django, MySQL, Python, HTML/CSS, JavaScript, Bootstrap

Book Recommendation System 03/03/2026 - 25/05/2026

- Developed a content-based NLP recommendation engine to generate personalized book suggestions.
- Data engineering (~10k books): data cleaning, missing-value handling, feature engineering (title, author, genres), and reliability filtering (>1,000 ratings).
- Modeling & ranking: TF-IDF, cosine similarity, and rating-based re-ranking.
- Optimization & deployment: model serialization with Pickle for fast and scalable execution.

Technologies : Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Pickle

AI Compiler 02/04/2025 - 30/05/2025

- Flexible multi-language execution: supports Python, Java, and C++ through file upload or direct input, with automatic language detection and instant results.
- AI-powered error assistance: provides clear error messages, simplified explanations, and intelligent correction suggestions.
- Intelligent analysis and learning: applies TF-IDF and lexical processing to error history to improve recommendations.

CERTIFICATIONS

- Introduction to Object-Oriented Programming in C++
- Interactivity with JavaScript — Coursera
- Successful Negotiation: Essential Strategies and Skills
- Using Python to Access Web Data
- React Native — Meta
- Software Engineering: Software Design and Project Management